



Data Science - Project Course

Lecture 2: Descriptive Statistics

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Topics of the day...

- Role of Statistics and Data Analysis
- Data summarization
- Concepts of Descriptive Statistics
- Outlier Detection
- Graphical summarization

Introduction

- We encounter data and make conclusions based on data every day.
- **Statistics** is the scientific discipline that provides methods to help us make sense of data.
- Statistical methods, used intelligently, offer a set of powerful tools for gaining insight into the world around us.
- The field of statistics teaches us how to make intelligent judgments and informed decisions in the presence of uncertainty and variation.

The Nature and Role of Variability

- Statistical methods allow us to collect, describe, analyse and draw conclusions from data.
- If we lived in a world where all measurements were identical for every individual, these tasks would be simple.

- **Example of No Variability:**

Imagine a population consisting of all students at a particular university. Suppose that *every* student was enrolled in the same number of courses, spent exactly the same amount of money on textbooks this semester, and favoured increasing student fees to support expanding library services. For this population, there is *no* variability in number of courses, amount spent on books, or student opinion on the fee increase.

- **Example of Variability:**

L1 1st semester score for your batch.

Definitions

- Population and Sample:

The entire collection of individuals or objects about which information is desired is called the **population** of interest.

A **sample** is a subset of the population, selected for study.

- Descriptive Statistics:

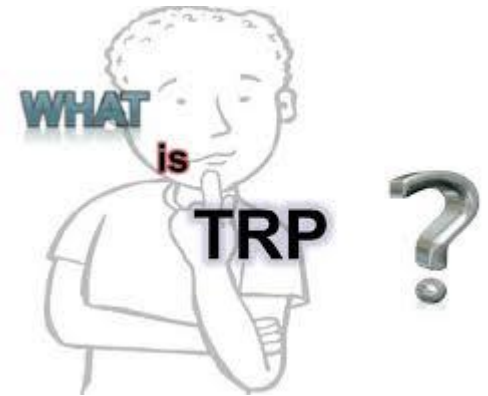
It is the branch of statistics that includes methods for organizing and summarizing data.

- Inferential Statistics:

It is the branch of statistics that involves generalizing from a sample to the population from which the sample was selected and assessing the reliability of such generalizations.

TRP: An example

- Television rating point (TRP) is a tool provided to judge which programs are viewed the most.
 - This gives us an index of the choice of the people and also the popularity of a particular channel.
- For calculation purpose, a device is attached to the TV sets **in few thousand** viewers' houses in different geographic and demographic sectors.
 - The device is called as **People's Meter**. It reads the time and the programme that a viewer watches on a particular day for a certain period.
- An average is taken, for example, for a 30-days period.
- The above further can be augmented with a personal interview survey (PIS), which becomes the basis for many studies/decision making.
- Essentially, we are to analyze **data** for TRP estimation.



Data

Definition : **Data**

A set of data is a collection of **observed values** representing one or more characteristics of some objects or **units**.

Example: For TRP, data collection consist of the following attributes.

- **Age:** A viewer's age in years
- **Sex:** A viewer's gender coded 1 for male and 0 for female
- **Happy:** A viewer's general happiness
 - NH for not too happy
 - PH for pretty happy
 - VH for very happy
- **TVHours:** The average number of hours a respondent watched TV during a day

Data : Example

Viewer#	Age	Sex	Happy	TVHours
...	...	...	...	...
...	...	...	...	...
55	34	F	VH	5
...	...	...	...	...

Note:

- A data set is composed of information from a set of units.
- Information from a unit is known as an observation.
- An observation consists of one or more pieces of information about a unit; these are called variables.

Type of Data

Variables:

A characteristic that varies from one person or thing to another is called a variable.

Example: height, weight, sex, marital status etc.

Quantitative (or Numerical) Variable:

A variable is numerical (or quantitative) if each observation is a number.

Example: height, weight etc.

Qualitative (or Categorical) Variable:

A variable is categorical (or qualitative) if the individual observations are categorical responses.

Example: sex, marital status etc.

Type of Data

Quantitative variable can also be classified as either discrete or continuous.

Discrete Variable:

A variable is discrete if it has only a countable number of distinct possible values i.e. a variable is discrete if it can assume only a finite numbers of values.

Example: Number of defects.

Continuous Variable:

A numerical variable is called continuous variables if the set of possible values forms an entire interval on the numerical line.

Example: Length, temperature etc.

Data: A collection of observations on one or more variables is called data.

Population

Definition : **Population**

A population is a data set representing the entire entities of interest.

Example: All TV Viewers in the country/world.

Note:

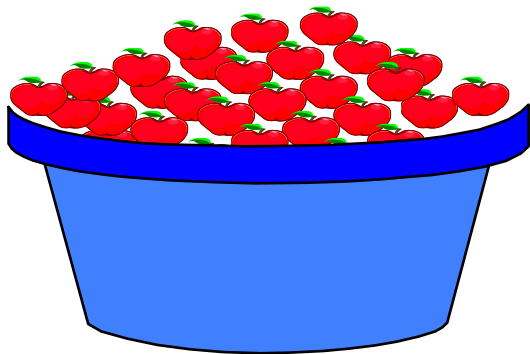
1. All people in the country/world is not a population.
2. For different survey, the population set may be completely different.
3. For statistical learning, it is important to define the population that we intend to study very carefully.

Sample

Definition : **Sample**

The small number of items taken from the population to make a judgment of the population is called a Sample. The numbers of samples taken to make this judgment is called *Sample size*.

Example: All students studying BSc Mathematics and Data Science in SUAD is a sample, whereas all students belong to SUAD is population.

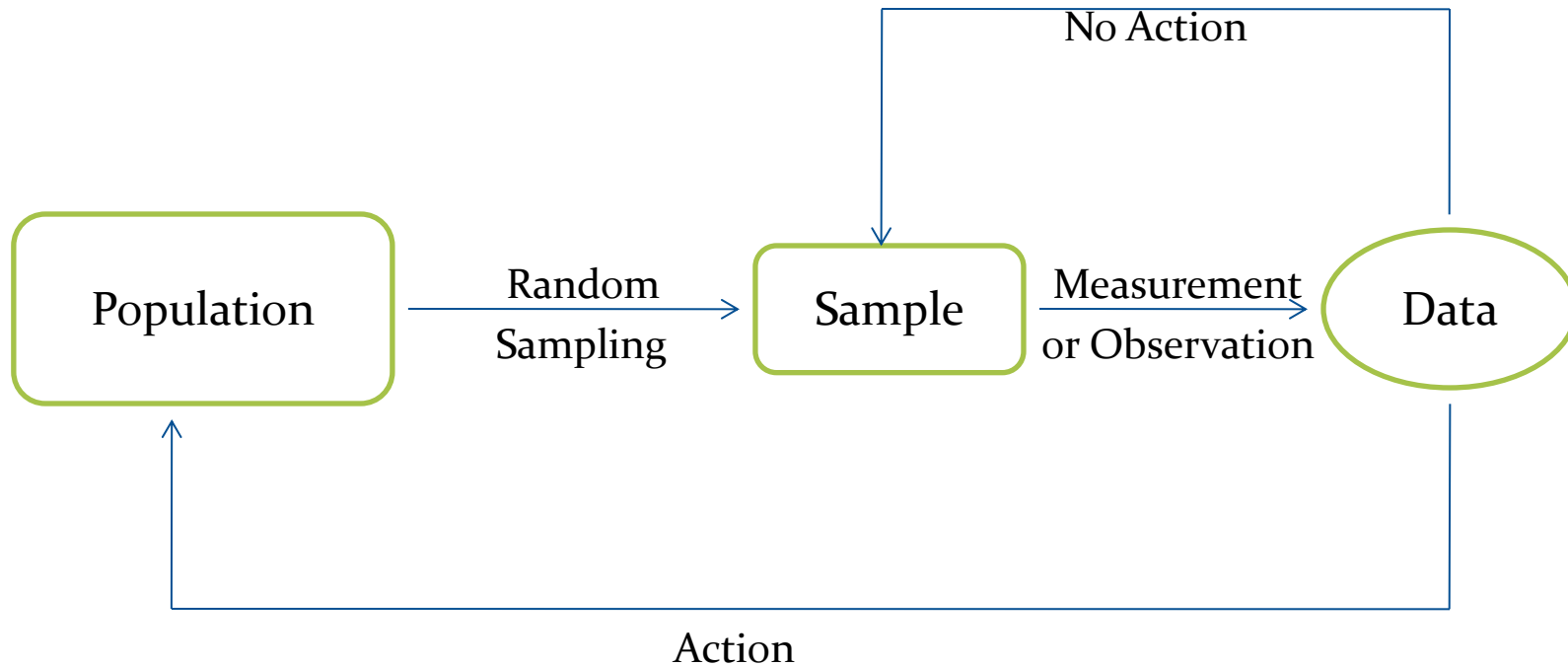


Population



Sample of size Three

Population, Sample and Data



Note: Normally a sample is obtained in such a way as to be representative of the population.

Statistic

Definition : Statistic

A statistic is a quantity calculated from data that describes a particular characteristics of a sample.

Example:

- The sample **mean** (denoted by $\bar{y}$) is the arithmetic mean of a variable of all the observations of a sample.
- **Statistic** (mean ($\bar{x}$), variance (s^2), etc.) consists of a body of methods for collecting analysing data.
- The probability distribution of a statistic Y is called the **sampling distribution** of Y .

Statistical Inference

Definition : Statistical inference

Statistical inference is the process of using sample statistic to make decisions about population.

Example: In the context of TRP

- Overall frequency of the various levels of happiness.
- Is there a relationship between the age of a viewers and his/her general happiness?
- Is there a relationship between the age of the viewer and the number of TV hours watched?

Data Summarization

- To identify the typical characteristics of data (i.e., to have an overall picture).
- To identify which data should be treated as noise or outliers.
- The data summarization techniques can be classified into two broad categories:
 - Measures of **location**
 - Measures of **dispersion**

Measurement of location

- It is also alternatively called as **measuring the central tendency**.
 - A function of the sample values that summarizes the location information into a single number is known as a measure of location.
- The most popular measures of location are
 - **Mean**
 - **Median**
 - **Mode**
- These can be measured in three ways
 - Distributive measure
 - Algebraic measure
 - Holistic measure

Distributive measure

- It is a measure (*i.e. function*) that can be computed for a given data set by partitioning the data into smaller subsets, computing the measure for each subset, and then merging the results in order to arrive at the measure's value for the original (*i.e. entire*) data set.

Example: `sum()`, `count()`

Algebraic measure

- It is a measure that can be computed by applying an algebraic function to one or more distributive measures.

Example: $\text{average} = \frac{\text{sum}()}{\text{count}()}$

Holistic measure

- It is a measure that must be computed on the entire data set as a whole.

Example: Calculating median. What about *mode*?

Mean of a sample

- The mean of a sample data is denoted as $\bar{x}$. Different mean measurements known are:
 - Simple mean
 - Weighted mean
 - Trimmed mean
- In the next few slides, we shall learn how to calculate the mean of a sample.
- We assume that given $x_1, x_2, x_3, \dots, x_n$ are the sample values.

Simple mean of a sample

- **Simple mean**

It is also called simply arithmetic mean or average and is abbreviated as (AM).

Definition : Simple mean

If $x_1, x_2, x_3, \dots, x_n$ are the sample values, the simple mean is defined as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Disadvantages of A.M

- It cannot be used if we are dealing with qualitative data.
- It cannot be obtained if a single observation is missing.
- It affected very much by extreme values.
- It may lead to wrong conclusions if the details of the data from which it is computed are not given.
 - Example: Let us consider the following marks obtained by two student A and B in three tests:

Marks	Test I	Test II	Test III	Average
A	50%	60%	70%	60%
B	70%	60%	50%	60%

Weighted mean of a sample

- **Weighted mean**

It is also called weighted arithmetic mean or weighted average.

Definition : **Weighted mean**

When each sample value x_i is associated with a weight w_i , for $i = 1, 2, \dots, n$, then it is defined as

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Note: *When all weights are equal, the weighted mean reduces to simple mean.*

Trimmed mean of a sample

- **Trimmed Mean**

If there are extreme values (*also called outlier*) in a sample, then the mean is influenced greatly by those values. To offset the effect caused by those extreme values, we can use the concept of trimmed mean

Definition : Trimmed mean

Trimmed mean is defined as the mean obtained after chopping off values at the high and low extremes.

Properties of mean

- Property 1:

If $\bar{x}_i$, $i = 1, 2, \dots, m$ are the means of m samples of sizes $n_1, n_2, \dots, n_m$ respectively, then the mean of the combined sample is given by:-

$$\bar{x} = \frac{\sum_{i=1}^m n_i \bar{x}_i}{\sum_{i=1}^m n_i}$$

(Distributive Measure)

- Property 2:

If m observations with mean $\bar{x}_m$, are added (*removed*) from a sample of size n with mean $\bar{x}_n$, then the new mean is given by

$$\bar{x} = \frac{n \bar{x}_n \pm m \bar{x}_m}{n \pm m}$$

Mean with grouped data

Sometimes data is given in the form of classes and frequency for each class.

<i>Class</i> →	$x_1 - x_2$	$x_2 - x_3$		$x_i - x_{i+1}$		$x_{n-1} - x_n$
<i>Frequency</i> →	f_1	f_2		f_i		f_n

There three methods to calculate the mean of such a grouped data.

- Direct method

Direct method

- Data Set: (Group Data)

<i>Marks:</i>	0 – 4	4 – 8	8 – 12	12 – 16	16 – 20
<i>No. of</i>					
<i>Student (f):</i>	2	18	10	20	10

- Direct Method

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

Where, $x_i = \frac{1}{2}$ (**lower limit + upper limit**) of the i^{th} class, i.e., $x_i = \frac{x_i + x_{i+1}}{2}$
(also called class size), and f_i is the frequency of the i^{th} class.

Some other measures of mean

- There are three mean measures of location:
 - Arithmetic Mean (AM)
 - Geometric mean (GM)
 - Harmonic mean (HM)

These three means are called **Pythagorean means**.

Some other measures of mean

- Arithmetic Mean (**AM**)

- $S: \{x_1, x_2\}$
- $\bar{x} = \frac{x_1 + x_2}{2}$
- $\bar{x} - x_1 = x_2 - \bar{x}$

- Geometric mean (**GM**)

- $S: \{x_1, x_2\}$
- $\tilde{x} = \sqrt{x_1 \cdot x_2}$
- $\frac{x_1}{\tilde{x}} = \frac{\tilde{x}}{x_2}$

- Harmonic Mean (**HM**)

- $S: \{x_1, x_2\}$
- $\hat{x} = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}}$
- $\frac{2}{\hat{x}} = \frac{1}{x_1} + \frac{1}{x_2}$

Geometric mean

Definition : Geometric mean

Geometric mean of n observations (*none of which are zero*) is defined as:

$$\tilde{x} = \left(\prod_{i=1}^n x_i \right)^{1/n}$$

where, $n \neq 0$

Note

- GM is the arithmetic mean in “log space”. This is because, alternatively,

$$\log \tilde{x} = \frac{1}{n} \sum_{i=1}^n \log x_i$$

- This summary of measurement is meaningful only when all observations are > 0 .
- If at least one observation is zero, the product will itself be zero! For a negative value, root is not real

Applications of GM

- The geometric mean is most useful to calculate the compounded growth rate where values in the sample are **not independent of each other**

Example: Calculation of growth rate

Consider a stock that grows by 10% in year one, declines by 20% in year two, and then grows by 30% in year three. What is the growth rate?

$$\begin{aligned} GM &= \sqrt[3]{(1 + 0.1)(1 - 0.2)(1 + 0.3)} \\ &= 0.046 \\ &= 4.6\% \text{ annually.} \end{aligned}$$

Harmonic mean

Definition : **Harmonic mean**

If all observations are non zero, the reciprocal of the arithmetic mean of the reciprocals of observations is known as harmonic mean.

For ungrouped data

$$\hat{x} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

For grouped data

$$\hat{x} = \frac{\sum_{i=1}^n f_i}{\sum_{i=1}^n \left(\frac{f_i}{x_i} \right)}$$

where, f_i is the frequency of the i^{th} class with x_i as the center value of the i^{th} class.

Applications of HM

- In fact, harmonic mean is the reciprocal of mean of sum of reciprocals
 - Harmonic means are often used in averaging things like rates.

Example: Calculate travel speed given durations of several trips

- A car in first 1 hour travels 60 kmph, in next 2 hours it travels with 90 kmph and next in 3 hours it travels with 80 kmph

Calculation using AM:

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{60+180+240}{1+2+3} = \frac{480}{6} = 80 \text{ kmph}$$

Calculation using HM

$$\text{Average speed} = \frac{1+2+3}{\frac{1}{60} + \frac{2}{90} + \frac{3}{80}} = \frac{6}{0.0167+0.0222+0.0375} = \frac{6}{0.0764} = 78.5 \text{ kmph}$$

Rule of thumbs for means

- For arithmetic mean
 - Use the **transformation** $y_i = x_i$
 - Take the arithmetic mean of all y_i s to get $\bar{y}$
 - Finally, $\bar{x} = \bar{y}$
- For geometric mean
 - Use the **transformation** $y_i = \log(x_i)$
 - Take the arithmetic mean of all y_i s to get $\bar{y}$
 - Finally, $\hat{x} = e^{\bar{y}}$
- For harmonic mean
 - Use the **transformation** $y_i = \frac{1}{x_i}$
 - Take the arithmetic mean of all y_i s to get $\bar{y}$
 - Finally, $\tilde{x} = \frac{1}{\bar{y}}$
- If your sample contains all values which are same (i.e., invariant), then

$$AM \geq GM \geq HM$$

Try to Prove it for n positive real numbers.

Exercise 2: Suppose there are three observations with $AM = 4$, $GM = 3.63$, $HM = 2.67$. Find the three observations?

Median

- Median of a distribution is the value of the variable which divides it into two equal parts.
- Median is not at all affected by extreme values.
- Ungrouped Data:
 - If the number of observations is odd then median is the middle value after the values have been arranged in ascending or descending order of magnitude.
 - In case of even number of observations, there are two middle values and median is obtained by taking the A.M of the middle values.

Median of a sample

Definition : **Median of a sample**

Median of a sample is the middle value when the data are arranged in increasing (*or decreasing*) order. Symbolically,

$$\hat{x} = \begin{cases} x_{(n+1)/2} & \text{if } n \text{ is odd} \\ \frac{1}{2} \{x_{n/2} + x_{(\frac{n}{2}+1)}\} & \text{if } n \text{ is even} \end{cases}$$

- Median is not at all affected by extreme values.

Median of a sample

Definition : Median of a grouped data

Median of a grouped data is given by

$$\hat{x} = l + \left\{ \frac{\frac{N}{2} - cf}{f} h \right\}$$

where h = width of the median class

$$N = \sum_{i=1}^n f_i$$

f_i is the frequency of the i^{th} class, and n is the total number of groups

cf = the cumulative frequency

N = the total number of samples

l = lower limit of the median class

Note

A class is called **median class** if its cumulative frequency is just greater than $N/2$

Mode of a sample

- Mode is defined as the observation which occurs most frequently.
- For example, number of wickets obtained by bowler in 10 test matches are as follows.

1 2 0 3 2 4 1 1 2 2

- In other words, the above data can be represented as:-

value	0	1	2	3	4
# of matches	1	3	4	1	1

- Clearly, the mode here is “2”.
- If a distribution has two modes, then it is called **bimodal**.

Mode of a grouped data

Definition : **Mode of a grouped data**

Select the modal class (it is the class with the highest frequency). Then the mode $\tilde{x}$ is given by:

$$\tilde{x} = l + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) h$$

where,

h is the class width

Δ_1 is the difference between the frequency of the modal class and the frequency of the class just after the modal class

Δ_2 is the difference between the frequency of the modal class and the class just before the modal class

l is the lower boundary of the modal class

Note

If each data value occurs only once, then there is no mode!

Relation between mean, median and mode

- There is an empirical relation, valid for moderately skewed data

$$\textit{Mean} - \textit{Mode} = 3 * (\textit{Mean} - \textit{Median})$$

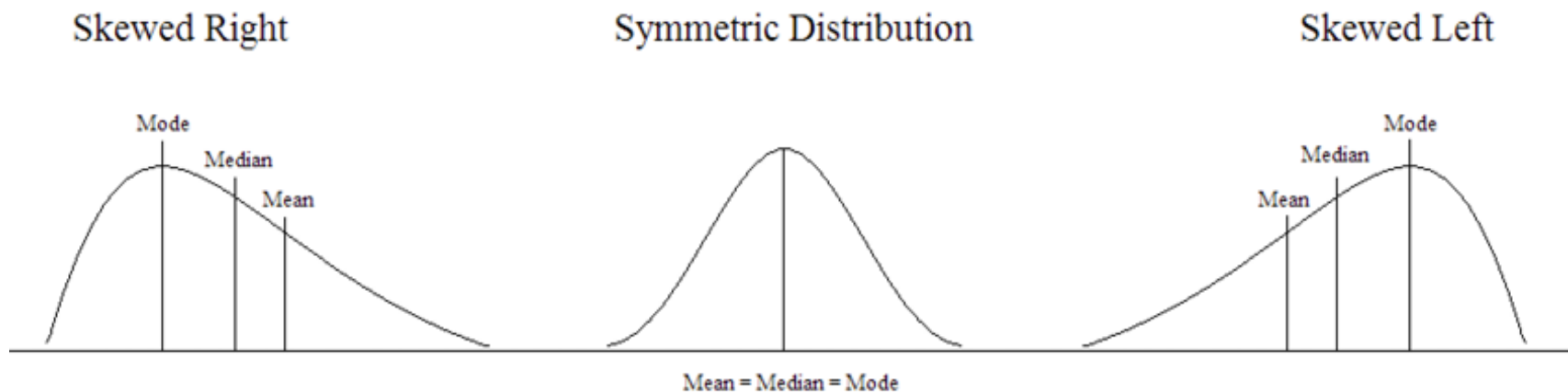
- A given set of data can be categorized into three categories:-
 - Symmetric data
 - Positively skewed data
 - Negatively skewed data
- To understand the above three categories, let us consider the following
- Given a set of m objects, where any object can take values $v_1, v_2, \dots, v_k$. Then, the frequency of a value v_i is defined as

$$\textit{Frequency}(v_i) = \frac{\textit{Number of objects with value } v_i}{n}$$

for $i = 1, 2, \dots, k$

Symmetric & Skewed data

- For symmetric data, all mean, median and mode lie at the same point.
- Positively Skewed Data: Mode occurs at a value smaller than the median.
- Negatively Skewed Data: Mode occurs at a value greater than the median.



Categorical Data

The **sample proportion of successes**, denoted by $\hat{p}$, is

$$\begin{aligned}\hat{p} &= \text{sample proportion of successes} \\ &= \frac{\text{number of } S\text{'s in the sample}}{n}\end{aligned}$$

where S is the label used for the response designated as success.

Measures of dispersion

- Location measure are far too insufficient to understand data.
- Another set of commonly used summary statistics for continuous data are those that measure the dispersion.
- A dispersion measures the extent of spread of observations in a sample.
- Some important measure of dispersion are:
 - Range
 - Variance and Standard Deviation
 - Interquartile Range (IQR)

Measures of dispersion

Example

- Suppose, two samples of fruit juice bottles from two companies *A* and *B*. The unit in each bottle is measured in litre.

Sample A	0.97	1.00	0.94	1.03	1.06
Sample B	1.06	1.01	0.88	0.91	1.14

- Both samples have same mean. However, the bottles from company A with more uniform content than company B.
- We say that the dispersion (or variability) of the observation from the average is less for A than sample B.
 - The variability in a sample should display how the observation spread out from the average
 - In buying juice, customer should feel more confident to buy it from A than B

Range of a sample

Definition : **Range of a sample**

Let $\mathbf{X} = \mathbf{x}_1, \dots, \mathbf{x}_n$ be n sample values that are arranged in increasing order.

The range $\mathbf{R}$ of these samples are then defined as:

$$\begin{aligned}\mathbf{R} &= \max(\mathbf{X}) - \min(\mathbf{X}) = \mathbf{x}_n - \mathbf{x}_1 \\ &= \text{Largest observation} - \text{Smallest observation}\end{aligned}$$

- Range identifies the maximum spread, it can be misleading if most of the values are concentrated in a narrow band of values, but there are also a relatively small number of more extreme values.
- The variance is another measure of dispersion to deal with such a situation.

Variance and Standard Deviation

Definition : Variance and Standard Deviation

Let $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ are sample values of n samples. Then, variance denoted as σ^2 is defined as :-

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{S_{xx}}{n-1}$$

where, $\bar{x}$ denotes the mean of the sample

The standard deviation, s , of the samples is the square root of the variance S^2

The **sample standard deviation** is the positive square root of the sample variance and is denoted by s .

Why $(n-1)$ is in the denominator instead of n ?

Variance and Standard Deviation

- **Lemma:** If data are transformed as $\mathbf{x}' = \frac{(\mathbf{x}-\mathbf{a})}{\mathbf{c}}$, the variance is transformed as

$$\sigma'^2 = \frac{1}{c^2} \sigma^2$$

Proof

The new mean $\bar{\mathbf{x}}' = \frac{\bar{\mathbf{x}}-\mathbf{a}}{\mathbf{c}}$

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (\mathbf{x}'_i - \bar{\mathbf{x}}')^2 &= \frac{1}{n} \sum_{i=1}^n \left\{ \frac{(\mathbf{x}_i - \mathbf{a})}{\mathbf{c}} - \frac{(\bar{\mathbf{x}} - \mathbf{a})}{\mathbf{c}} \right\}^2 \\ &= \frac{1}{c^2 n} \sum_{i=1}^n \{ (\mathbf{x}_i - \mathbf{a}) - (\bar{\mathbf{x}} - \mathbf{a}) \}^2 \\ &= \frac{1}{c^2 n} \sum_{i=1}^n \{ \mathbf{x}_i - \bar{\mathbf{x}} \}^2 \\ &= \frac{1}{c^2} \sigma^2 \quad \text{[PROVED]} \end{aligned}$$

Coefficient of Variation

- **Basic properties**

- σ measures spread about mean and should be chosen only when the mean is chosen as the measure of central tendency
- $\sigma = 0$ only when there is no spread, that is, when all observations have the same value, otherwise $\sigma > 0$

Definition : **Coefficient of variation**

A related measure is the coefficient of variation **CV**, which is defined as follows

$$\mathbf{CV = \frac{\sigma}{\bar{x}} \times 100}$$

This gives a ratio measure to spread.

Coefficient variation

- **Significance of CV**

- It is a statistical measure of the dispersion of data points in a data series around the mean
- **CV = p%** implies that standard deviation is p% to that of the mean of a sample.

Example:

- Suppose, there are three series of data representing amount of returns that investors receives from three farms F1, F2 and F3 in a year.
- CV(F1), CV(F2), CV(F3) indicate volatilities/ risks in comparison of return expected from investment

$$CV = \frac{\text{Volatility}}{\text{Expected Return}} \times 100$$

Coefficient variation

Mr. X is looking for a safe investment that provides safe and stable returns. There are following options:

- a) **Stocks:** The volatility of the stock is 10% and the expected return is 14%.
- b) **Mutual funds:** It offers an expected return of 13% with a volatility of 7%.
- c) **Fixed deposits:** This scheme offers an expected return of 3% with 2% volatility.

In order to select the most suitable investment opportunity, Mr. X should choose which investment scheme?

Interquartile Range

- **Quartile**

- The most commonly used percentiles are quartiles.
 - The first quartile, denoted by Q_1 is the 25th percentile.
 - The third quartile, denoted by Q_3 is the 75th percentile
 - The median, Q_2 is the 50th percentile.
- The quartiles including median, give some indication of the center, spread and shape of a distribution.
- The distance between Q_1 and Q_3 is a simple measure of spread that gives the range covered by the middle half of the data. This distance is called the interquartile range (**IQR**) and is defined as

$$\mathbf{IQR = Q_3 - Q_1}$$

Application of IQR

- **Outlier detection using five-number summary**
 - A common rule of the thumb for identifying suspected outliers is to single out values falling at least $1.5 \times \text{IQR}$ above Q_3 and below Q_1 .
 - In other words, extreme observations occurring within $1.5 \times \text{IQR}$ of the quartiles

Application of IQR

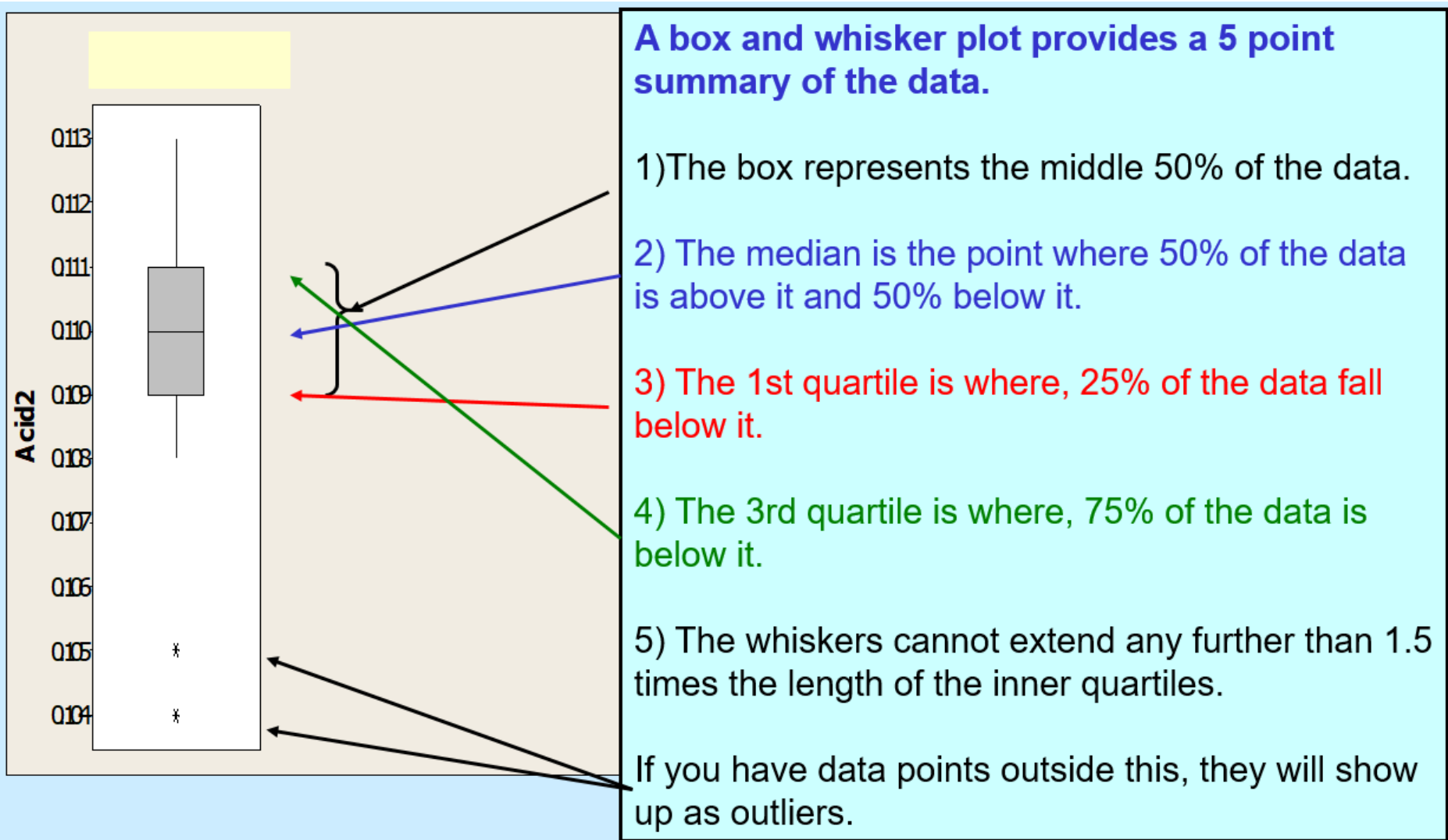
- **Five Number Summary**

- Since, Q_1 , Q_2 and Q_3 together contain no information about the endpoints of the data, a **complete** summary of the shape of a distribution can be obtained by providing the lowest and highest data value as well. This is known as the five-number summary
- The five-number summary of a distribution consists of :
 - The Median Q_2
 - The first quartile Q_1
 - The third quartile Q_3
 - The smallest observation
 - The largest observation

These are, when written in order gives the **five-number summary**:

Minimum, Q_1 , Median (Q_2), Q_3 , Maximum

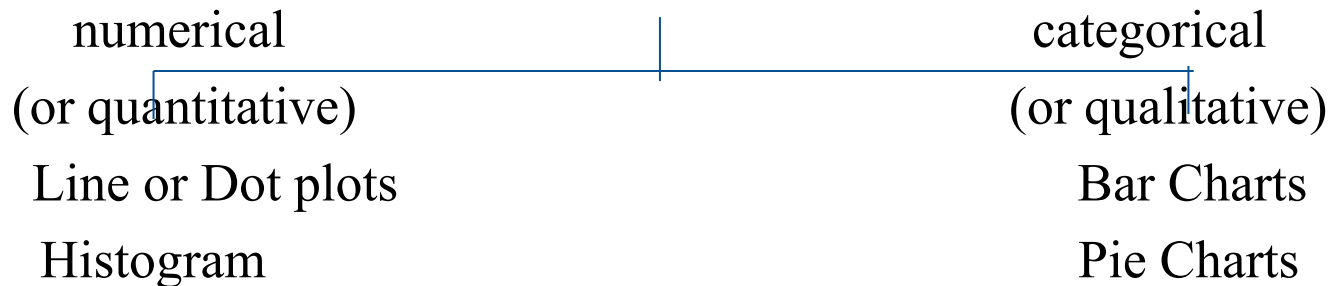
Box plot



Graphical Representation of Data

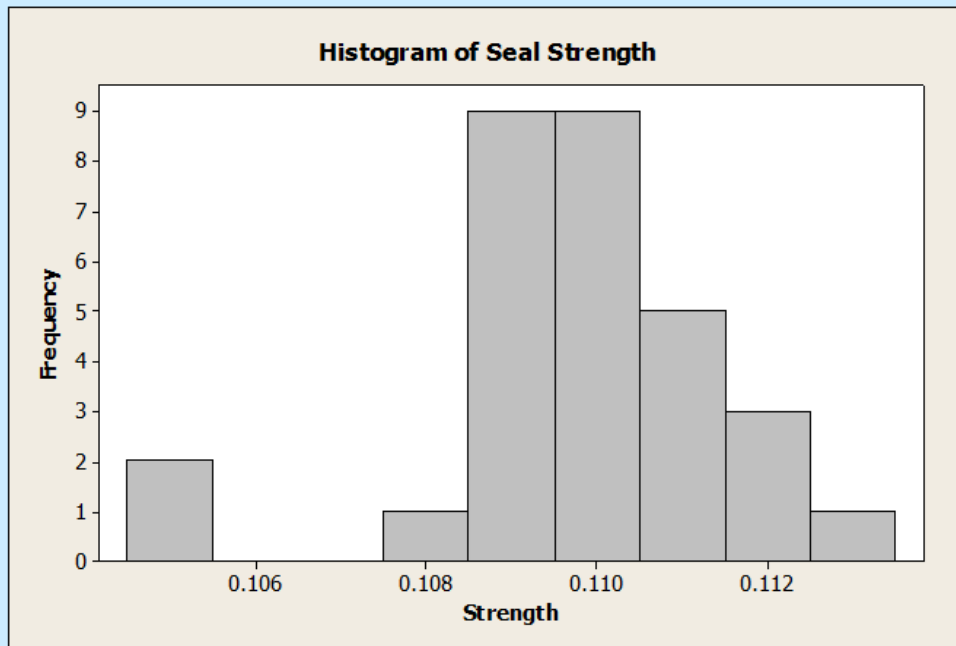
- Visualization techniques are ways of creating and manipulating graphical representations of data.
- We use these representations in order to gain better insight and understanding of the problem we are studying - pictures can convey an overall message much better than a list of numbers.

Data



Histogram

Histogram is a basic graphing tool that displays the relative frequency or occurrence of continuous data values showing which values occur most and least frequently.



A histogram illustrates the

Shape,
Centering, and
Spread

of data distribution

and indicates whether
there are any outliers.